Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/07/2021
PI Name:	Koochesfahani, Manoochehr Ph.D.		
Project Title:	ZBOT Thermometry and Velocimetry Diagnostics Development		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	koochesf@egr.msu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(517) 353-5311
Organization Name:	Michigan State University		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	428 S. Shaw Lane, Room 2555		
PI Web Page:			
City:	E. Lansing	**State:**	MI
Zip Code:	48824	**Congressional District:**	8
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Physical Sciences Unsolicited
Start Date:	02/05/2016	**End Date:**	12/31/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 12/31/2019 per NSSC; original end date was 2/14/2018 and had been extended to 7/14/2018 in Feb 2018 and then to 12/31/2018 in June 2018 (Ed., 3/28/19) NOTE: End date changed to 12/31/2018 per NSSC; original end date was 2/14/2018 and had been extended to 7/14/2018 in Feb 2018 (Ed., 7/31/18)		
Key Personnel Changes/Previous PI:	April 2019 report: Dr. Shahram Pouya moved to Iowa State University in 2018 and is no longer CoInvestigator on the project.		
COI Name (Institution):			
Grant/Contract No.:	NNX16AD52A		
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Performance Goal Text:			

Task Description:	In-orbit or on-surface cryogenic storage for propellant and life support consumables is essential to all stages of NASA's Exploration Program. However, heat leakage from the surroundings results in liquid vaporization and significant mass loss and/or significant increase in the storage tank pressure. To mitigate the unwanted effects of cryogen vaporization, the Zero Boil-Off (ZBO) concept is being considered as an innovative approach for ventless means of controlling tank pressure through the application of active heat removal and forced liquid jet mixing. The Zero Boil-Off Tank (ZBOT) experiment is designed for operation on the International Space Station (ISS). Its purpose is to examine the ZBO concept and, along with computational modeling efforts, provide information on the fundamental physics of the underlying complex thermofluids environment, which is needed for engineering design optimization of the ZBO concept. The ZBOT experiment currently relies on Digital Particle Image Velocimetry (DPIV) to obtain flow field information and visualization in the test fluid, Perfluoro-n-Pentane (PnP). Difficulties have been encountered, however, in using DPIV. These difficulties include a drastic loss of particles over time and decrease of fluorescence intensity over time from the fluorescent-coated particles, along with concerns over particle sticking and agglomeration. The proposed effort is motivated by the need to obtain non-intrusive, whole-field, temperature and/or velocity data using optical diagnostics in the ZBOT microgravity experiment in order to supplement the computational modeling efforts in progress. Because of the challenges experienced by the ZBOT team in using particle-based techniques (i.e., DPIV), our focus will be on development of molecular-based approaches for measurements within PnP. These include laser induced fluorescence (LIF) methods for thermometry and molecular tagging approaches for velocimetry (MTV).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. Introduction - This section describes our progress with initial ZBOT investigations. Our research has demonstrated the feasibility of molecular tagging velocimetry (MTV) – as well as laser induced fluorescence (LIF) thermometry – based on solubilized quantum dots (QD). Immediately following the list of specific progress areas is a conclusion with recommendations for future work and the path forward. 2. Tracers for Velocimetry and Thermometry - Exploring suitable molecular-based temperature and/or velocity measurement methods for the ZBOT microgravity experiment. This project was originally launched with the objective of developing non-intrusive molecular-based temperature and/or velocity measurement methods suitable for the ZBOT microgravity experiment. Our desire to develop these measurement methods was partly motivated by the potential challenges connected to the use of particle-based techniques, i.e., Digital Particle Image Velocimetry (DPIV). [Ed. Note December 2021: As noted in the FY19 NASA Task Book report, researchers processed particle image sequences obtained in the ISS ZBOT experiments in order to obtain quantitative information about the flow field. Both the particle streak mode of data interpretation and the DPIV mode of data processing were evaluated.] In addition, whole-field thermometry capability for the ZBOT experiment did not exist when this task was initiated. Our search for compounds with fluorinated functionality that would dissolve in PnP led us to CdSe/ZnS core-shell nanocrystal quantum dots (QD) that have been suitably functionalized. Taking a clue from the fluorous group that renders Coumarin 153 soluble in PnP, we successfully modified commercially available QDs to dissolve into PnP with the help of Professor Gary Blanchard - Michigan State University Department of Chemistry. The measured photochemical properties of QDs in PnP show promising temperature sensitivity (nearly three times higher than Coumarin 153) and a photobleaching rate that is 100 times less than other compounds. Calibration experiments have shown that a two-color LIF thermometry approach is feasible as a ratiometric technique. We therefore utilized QDs as the main thermometry tracers in subsequent ZBOT experiments. 3. Molecular Tagging Velocimetry (MTV) - Identification / synthesis of molecular tracers. A major challenge in this project was the identification and/or synthesis of molecular tracers that would properly dissolve in the fluid medium utilized in the ZBOT investigation. The fluid medium used in the ZBOT experiment was Perfluoro-n-Pentane (PnP). We characterized the photochemical properties of identified tracers to ensure that the tracers could be possible targets for either MTV, or LIF thermometry. Proof-of-concept experiments were carried out to ascertain feasibility of quantitative measurements. The proof-of-concept experiments were followed by demonstration experiments in the ZBOT breadboard setup, which has geometric characteristics similar to the actual ZBOT test section used on the International Space Station (ISS). Our investigation has shown that implementation of MTV, based on photobleaching of Coumarin 153, is feasible in the ZBOT breadboard setup; and MTV is accurately able to resolve the range of velocities of interest in the ZBOT experiment. Note that before the MTV measurements were conducted, a correction was made to mitigate the effect of test section curvature and mismatch of refractive indices. The research team imaged a calibration target inside the test tank to quantify the necessary correction. 4. LIF Thermometry with Quantum Dots We had previously demonstrated the promise of ratiometric thermometry in the ZBOT experiment by two-wavelength LIF measurement using modified QDs. To increase the accuracy of thermometry measurement, we desired a higher level of sensitivity. We were able to improve ratiometric sensitivity by using an optimized filter set. The transmission curves of the new filters, along with the emission spectrum of QDs at room temperature, resulted in a ratiometric sensitivity of 1.1%/°C (a 30% improvement over previous results). 5. Recommendations for Path Forward and Future Work While preliminary results are quite promising, they also point to additional work that is needed in the future to make the diagnostics robust and ready for implementation into ZBOT ISS hardware. Below are areas that will be explored in future work: A. Synthesis of Quantum Dots and Other Tracers with Fluorinated Functionality B. Molecular Tagging Velocimetry C. LIF Thermometry

	(Ed. note December 2021: compiled from PI's technical report submitted December 2019)
Bibliography Type:	Description: (Last Updated: 03/02/2026)
Articles in Peer-reviewed Journals	Pouya S, Blanchard G, Koochesfahani M. "Development of molecular tagging velocimetry for the ZBOT experiment." Exp. Fluids. 2019 Apr 9;60:78. https://doi.org/10.1007/s00348-019-2726-1 , Apr-2019